

Industry Problem Solving Task

Big Data – Questions and Answers

CO4-AT2

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1. Retail Data Optimization Problem

Question: A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Answer:

Big Data Architecture:

1. Data Sources – Collect data from POS systems, online orders, inventory systems, customer interactions, mobile applications, and websites.
2. Data Ingestion – Use Apache Kafka for real-time collection of sales, inventory, and customer events. Batch data can be loaded using ETL tools.
3. Data Storage – Store raw and historical data in a cloud data lake such as Amazon S3 or Azure Data Lake. A data warehouse can be used for structured reporting.
4. Stream Processing – Use Apache Spark Streaming or Apache Flink to process inventory and sales events in near real time.
5. Data Processing – Apache Spark can clean, transform, join, and aggregate large datasets.
6. Analytics Layer – Use dashboards such as Power BI or Tableau for inventory monitoring and business reporting.
7. Machine Learning – Use time-series forecasting models such as ARIMA, Prophet, or LSTM to predict product demand. Recommendation models such as collaborative filtering can provide personalized product recommendations.
8. Output – Store managers receive real-time inventory alerts, while customers receive personalized offers and recommendations.

Overall flow:

POS/Website/App → Kafka → Spark/Flink → Data Lake/Warehouse → ML Models →
Dashboards/Alerts/Recommendations.

2. Social Media Fraud Detection Problem

Question: A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Answer:

Big Data Solution:

1. Data Sources – Collect posts, comments, likes, shares, login activity, account information, and network/interaction data.
2. Real-Time Ingestion – Apache Kafka can capture billions of events continuously and distribute them to processing systems.
3. Stream Processing – Apache Flink or Spark Structured Streaming can process events in real time or near real time.
4. Data Storage – Use a distributed data lake such as HDFS or cloud object storage for historical data. NoSQL databases can store high-volume user and interaction data.
5. Feature Engineering – Generate features such as posting frequency, account age, IP/device patterns, unusual behavior, sharing patterns, and user-network relationships.
6. Machine Learning – Classification models such as Random Forest, Gradient Boosting, or neural networks can identify suspicious accounts. NLP models can analyze text for misinformation indicators. Graph-based models can identify suspicious relationships between accounts.
7. Real-Time Decision Layer – Suspicious activity can be assigned a risk score. High-risk accounts or posts can be flagged for review or automatically restricted.
8. Monitoring – Dashboards can show fraud trends, detected accounts, and misinformation alerts.

Overall flow:

Social Media Events → Kafka → Flink/Spark Streaming → Feature Extraction → ML Models → Risk Score →
Alert/Action → Historical Data Lake.

3. Government Data Platform Problem

Question: A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Answer:

Governance and Data Management Strategy:

1. Data Lake – Create a centralized data lake to store census, economic, and geospatial datasets in their original and processed forms.
2. Data Integration – Use ETL/ELT pipelines and common data formats/APIs to integrate datasets from different departments.
3. Metadata Management – Maintain a central data catalog containing dataset descriptions, owners, formats, quality information, and usage rules.
4. Interoperability – Define common schemas, identifiers, APIs, and standards so different government systems can exchange data consistently.
5. Data Quality – Apply validation, deduplication, completeness checks, and consistency rules.
6. Governance – Establish data owners, data stewards, access policies, retention rules, and data classification standards.
7. Security – Use encryption at rest and in transit, role-based access control, multi-factor authentication, audit logs, and network security controls.
8. Privacy – Protect personally identifiable information using masking, anonymization/pseudonymization, minimum-access principles, and controlled data sharing.
9. Monitoring and Compliance – Continuously monitor access, data usage, security events, and compliance requirements.

Result:

A governed national data platform provides integrated and reusable data while maintaining interoperability, privacy, security, quality, and accountability.

4. Banking Fraud Detection Problem

Question: A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Answer:

Big Data Pipeline:

1. Data Sources – Collect transaction records, customer profiles, account history, device information, location, and previous fraud records.
2. Streaming Ingestion – Use Apache Kafka to receive transactions continuously with high throughput.
3. Stream Processing – Use Apache Flink or Spark Structured Streaming to process transactions with low latency.
4. Historical Storage – Store transaction history in a distributed data lake and use a warehouse/database for analytical queries.
5. Feature Engineering – Calculate transaction amount, transaction frequency, location changes, unusual time patterns, merchant information, and customer spending behavior.
6. Machine Learning – Supervised models such as Logistic Regression, Random Forest, XGBoost, or neural networks can classify transactions as legitimate or fraudulent. Anomaly detection can identify previously unseen fraud patterns.
7. Real-Time Scoring – Each transaction is passed to the ML model and receives a fraud probability/risk score.
8. Decision Layer – High-risk transactions can be blocked or sent for verification; low-risk transactions can be approved.
9. Feedback – Confirmed fraud cases are stored and used to retrain and improve the models.

Overall flow:

Transactions → Kafka → Flink/Spark → Feature Engineering → ML Model → Fraud Score →
Approve/Block/Review → Data Lake → Model Retraining.

5. Agriculture Smart Farming Problem

Question: A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Answer:

Big Data Solution:

1. Data Sources – Collect soil moisture, temperature, humidity, rainfall, crop health, satellite/drone images, irrigation data, and weather information.
2. IoT Data Collection – Sensors installed across fields continuously generate data. Edge devices can collect and preprocess data close to the farm.
3. Data Ingestion – Use MQTT/IoT gateways and Kafka to transfer sensor and drone data to the cloud.
4. Storage – Store large-scale sensor, image, and historical data in a cloud data lake.
5. Stream Processing – Spark Streaming or Flink can process real-time sensor readings and generate alerts.
6. Analytics – Descriptive analytics shows current soil and crop conditions. Predictive analytics can forecast crop yield, irrigation needs, pest risks, and weather impacts.
7. Machine Learning – Regression models can predict yield and water requirements. Classification/deep learning models can detect crop diseases from drone or image data.
8. Decision Support – The system can recommend when to irrigate, how much water/fertilizer to use, and where crop problems require attention.
9. Visualization – Dashboards and mobile applications can display field maps, alerts, crop health, and recommendations.

Overall flow:

Sensors/Drones/Weather → IoT Gateway → Kafka → Cloud Data Lake → Spark/Flink → ML & Analytics → Alerts/Recommendations → Farmer Dashboard.